

- **Surface anatomy:**

- * **Elevations**

- The labial surface of the crown is convex with the maximum convexity at the cervical third (***cervical ridge***).
 - **Labial ridge** is prominent extending from the cusp tip to the cervical ridge.

- * **Depressions**

There are two shallow developmental grooves mesial and distal to the labial ridge, which divide the labial surface into three lobes.

- 2- **The root:**

This tooth has a single root. It is long and slender. The apex is pointed and distally curved.

Lingual aspect: (Fig. 35)

- 1- **The crown:**

The geometric outline and outlines form are similar to that of the labial aspect. However it differs in the **crown size** and **surface anatomy**.

- The crown is narrower lingually than labially (lingual convergence).
- Similar to incisors, cingulum and marginal ridges (mesial and distal) are characteristics for the lingual surfaces of anterior teeth.
- Lingual ridge: a prominent vertical ridge extends from the cusp tip to the cingulum. Canines are the only teeth that have a prominent lingual ridge.
- Lingual fossae: on each side of the lingual ridge a mesial and distal lingual fossae are present.



Fig. 35:
Lingual aspect.

The root:

The lingual surface of the root is smooth and convex. It is narrower lingually than labially.

Maxillary Canine (Fig. 33)

Labial aspect: (Fig. 34)

For proper description of the labial aspect of canines;

First, describe the crown:

- a. *Geometric outline.*
- b. *Outline form; proximal (mesial and distal with reference to the contact areas), cervical and incisal outlines (with reference to the cusp slopes, shape and position of the cusp tip).*
- c. *Surface anatomy; elevations and depressions.*

Second, describe the root:

Number, shape and apex (position and shape).

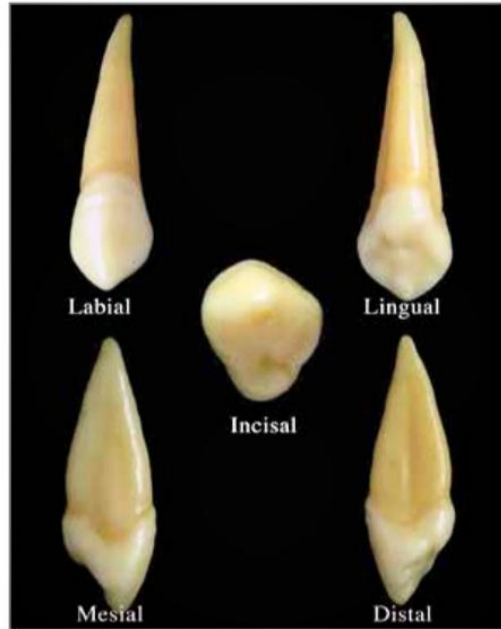


Fig. 33: Maxillary left permanent canine.

1- The crown:

- The **geometric outline** of the crown is **trapezoidal** with the shortest of the uneven sides is located **cervically**.
- **The outline form:**
 - * **The mesial outline:** from the cervical line to the contact area is **convex**. The **contact area** is nearly at the junction of the middle and incisal thirds of the crown.
 - * **The distal outline:** from the cervical line to the contact area is concave. The **contact area** is more cervically than mesial (in the middle third).
 - * **The incisal outline:** The **mesial cusp slope** is **concave** while the **distal cusp slope** is **convex** and **longer** than the mesial. The cusp tip is pointed and mesial to the midline (long distal slope).
 - * **The cervical outline:** convex root-wards.



Fig. 34: Labial aspect.

CANINES

General characteristics: (Fig. 32)

- There are one canine in each quadrant.
- Canines have **longer and stronger roots** than those of other teeth.
- Over the labial aspect of their roots there is well developed bone ridge called the **canine eminence** giving the canine a very important cosmetic value since they ensure facial expression.
- Canines are formed of four lobes, (3 labial and 1 lingual), the middle labial lobe is highly developed labially (forming the labial ridge), lingually (forming the lingual ridge) and incisally (forming a well developed pointed cusp). Therefore, the canines are called **cuspids**.
- The canines are located between the incisors and premolars, so, they support the two groups in **function** (tearing and partial cutting).



Fig. 32: The upper and lower permanent canines.

Mesial aspect: (Fig. 36)

For proper description of the proximal aspects of canines;

First-Describe the crown:

- a. Geometric outline.
- b. Outline form; labial, lingual (with reference to the position of the cusp tip) and cervical.
- c. Position of the contact area.

Second-Describe the root:

Number, shape and apex (position and shape).



Fig. 36:
Mesial aspect.

1- **The crown:**

- **The geometric outline** is triangular in shape with the base of the triangle at the cervix and the apex at the cusp tip.
- **The outline form:**
 - * ***The labial outline:*** is **convex** with the maximum convexity is at the cervical third (***cervical ridge***).
 - * ***The lingual outline:*** is **convex** in the cervical third (***cingulum***), **straight** in the middle third (***lingual ridge***), and **convex** in the incisal third representing the cusp tip.
 - * ***The cervical line:*** curves incisally to a marked degree.

Note; The cusp tip is located labially to the root axis.

- **The contact area:**

It is **centralized** (labio-lingually) and nearly **at the junction of the middle and incisal thirds** of the crown (inciso-cervically).

2-**The root:**

- The single root is **conical**. Its apex is **blunt and curved labially**.
- The mesial surface is broad with a shallow longitudinal depression.

Distal aspect: (Fig. 37)

- The geometric outline and outlines form are similar to that of the mesial aspect, except;
- The cervical line has less curvature toward the cusp tip.
- The contact area is still centralized but is located more cervically.
- The depression on the distal surface of the root is more pronounced than that on the mesial surface. These vertical depressions on the heavy root help to anchor the teeth in their socket bone and help to prevent rotation and displacement.



Fig. 37: Distal aspect.

Incisal aspect: (Fig. 38)**For proper description of the incisal aspects of canines;**

Similar to that of the incisors in addition to the position of the cusp tip

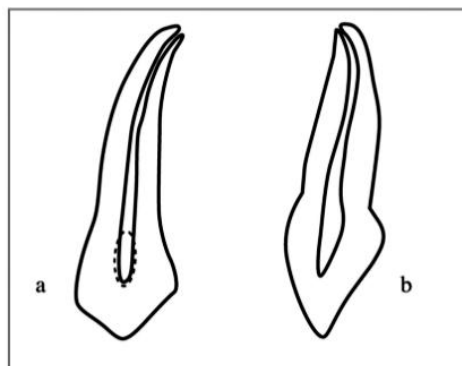
- The crown outline is diamond.
 - * The crown converges lingually.
 - * The crown thickness is larger than its width.
- The labial and lingual surfaces are convex, more than either incisor.
- The **cusp tip** is labial and mesial to the centers of the crown.
- The **labial ridge** is very noticeable from this aspect.
- The **lingual ridge** extends from the cusp tip to the cingulum in between the two lingual fossae.
- The **cingulum and marginal ridges** are also seen from this aspect.

Fig. 38:
Incisal aspect.**Pulp cavity:** (Fig. 39)*** Mesiodistally** (Fig.39 a);

- The pulp cavity is much narrower in the mesiodistal section than in the incisors.
- The pulp chamber is pointed incisally with a prominent pulp horn corresponding to the cusp.
- The root canal is narrow along its entire length.

*** Labiolingually** (Fig.39 b);

- Wide root canal and pulp chamber with pointed pulp horn.

Fig. 39: Pulp cavity of upper permanent canine.
a, mesiodistal section. b, labiolingual section.

Mandibular Canine (Fig. 40)

- It closely resembles the upper canine.
- Its anatomical landmarks are less developed than those of the upper canine.

Labial aspect: (Fig. 41)

- It resembles that of the upper canine with some differences:
- Its width is narrower than the upper canine but longer.
- The mesial outline is nearly straight from the cervical line to the mesial contact area.
- The mesial contact area is in the incisal third while the distal is at the junction of the middle and incisal thirds, more cervically located than the mesial contact area.

Note: The crown of the lower canine appears longer than that of the upper canine. Sometimes it is longer, but the effect of greater length is emphasized by the narrowness of the crown and incisally positioned contact areas.

- The cusp is less pointed than that of upper canine.
- The anatomical landmarks are less developed than those of the upper canine.
- The root is shorter than that of the upper canine with more pointed apex.

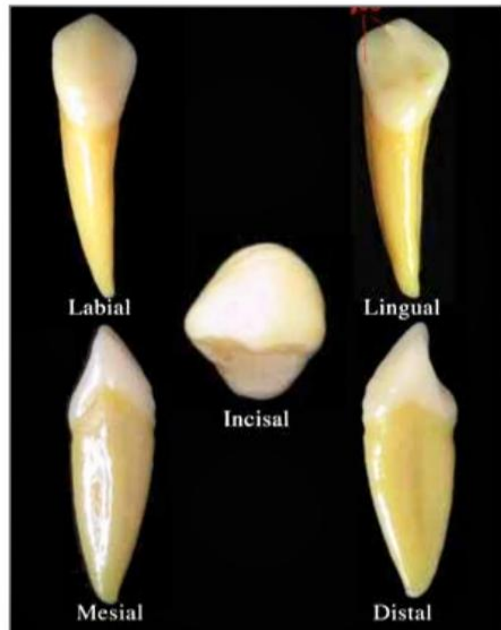


Fig. 40: Mandibular left permanent canine.



Fig. 41: Labial aspect.

Lingual aspect: (Fig. 42)

- It is similar to that of the upper canine with some differences;
 - * The lingual ridge is restricted to the incisal third leading to presence of one lingual fossa.
 - * The cingulum and marginal ridges are not prominent



Fig. 42:
Lingual aspect.

Mesial aspect: (Fig. 43)

- The mesial aspect of the lower canine is similar to that of the upper canine with some differences:
- The convexity of the labial outline is less prominent due to less developed anatomical landmarks.
- The lingual outline in the middle third is concave due to absence of long prominent lingual ridge of the upper canine.
- The cusp tip is pointed and located lingual to the root axis line (lingual inclination) or may be centered over it.
- The contact area is centralized and in the incisal third.
- The developmental depression on the root mesially is more pronounced than that on the upper canine root.



Fig. 43: Mesial aspect.

Distal aspect: (Fig. 44)

- The curvature of the cervical line is less in depth than on the mesial side.
- The distal contact area is still centralized but is located more cervically.
- The developmental depression on the distal side of the root is deeper than the depression on the mesial side.



Fig. 44:
Distal aspect.

Note; The developmental depression on the root in rare cases may be deep that divide the root into two labial and lingual roots (Fig. 45).

Incisal aspect: (Fig. 46)

- The outline of lower canine from the incisal aspect is similar to that of the upper canine with some differences;
- The cusp tip inclined lingually or centralized.
- The lingual ridge is restricted to the incisal third.



Fig. 46:
Incisal aspect.



Fig. 45: Lower canine
with bifurcated root.

Pulp cavity:

- The pulp cavity of lower canine resembles that of the upper canine except the possibility of presence of two separate labial and lingual root canals.

**Chronology of the permanent anterior teeth
is summarized in the following table.**

	Beginning of calcification	Crown completion	Eruption	Root Completion
$\frac{1}{1}$	3-4 m.	3 y. before eruption date	7 y.	3 y. after eruption date
			6 y.	
$\frac{2}{2}$	10-12 m. 3-4 m.		8 y.	
			7 y.	
$\frac{3}{3}$	4-5 m		11 y.	
			9 y.	